Dynamic Spectral Asset Pricing and the Sovereign Hierarchy Index

Theory, Large Deviations, and GMM Estimation

Soumadeep Ghosh

Kolkata, India

Abstract

We develop a dynamic spectral asset pricing framework for international equity markets and introduce the *Dynamic Sovereign Hierarchy Index* (DSHI), a scalar measure of eigenvalue concentration in conditional covariance matrices of excess returns. The framework rests on four interconnected pillars. First, we derive a spectral stochastic discount factor (SDF) whose factor loadings are identified directly from the eigenstructure of the time-varying covariance matrix, and we establish the associated no-arbitrage pricing restrictions. Second, we model eigenvalue dynamics as a multivariate square-root (CIR-type) diffusion and characterise the stationary distribution of the DSHI via its cumulant generating function. Third, we derive a large-deviation principle for the DSHI using the Gärtner–Ellis theorem and obtain saddlepoint approximations for both the marginal and joint densities of hierarchy and volatility. Fourth, we embed the spectral SDF in a full system GMM framework, derive the optimal weighting matrix, closed-form estimator, and asymptotic variance, and provide a χ^2 over-identification test. Simulation-calibrated results indicate that the global spectral mode carries an economically and statistically significant risk premium, whereas sub-bloc modes contribute modestly and heterogeneously. The framework accommodates regime-switching, cross-sectional dependence, and latent factor instability, making it well suited for monitoring evolving patterns of financial globalisation.

The paper ends with "The End"

Keywords: spectral asset pricing; dynamic factor models; sovereign hierarchy; GMM; large-deviation theory; saddlepoint approximation; random matrix theory; international equity markets.

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1 Introduction

The architecture of global financial markets has shifted markedly over the past three decades. Cross-country return correlations have risen, co-movement clusters have reorganised around economic blocs, and the dominance of a single global factor has at times given way to more diffuse, multi-polar risk structures. Understanding these dynamics is both empirically important and theoretically under-explored.

Existing international asset pricing models—from the early works of Solnik (1974) and Adler and Dumas (1983) to the stochastic-discount-factor formulations of Cochrane (2005)—focus primarily on the *number* of priced risk factors. They say comparatively little about the *geometry* of those factors: how concentrated is risk in a single dominant direction, and how does that concentration evolve over time?

This paper addresses the gap. We construct a *dynamic spectral asset pricing* (DSAP) framework in which the pricing factors are the principal components of the conditional covariance matrix Σ_t of excess returns, and we introduce the *Dynamic Sovereign Hierarchy Index* (DSHI) as a normalised Herfindahl measure of eigenvalue concentration. The DSHI equals one when a single global factor explains all cross-country variance, and equals zero when risk is uniformly distributed across N orthogonal directions.

Our contributions are as follows.

1. **Spectral SDF.** We derive a no-arbitrage SDF $m_t = 1-\theta' F_t$ whose spectral factors $F_t = Q'_{t-1} R^e_t$ are identified from the eigendecomposition of Σ_t. We establish the duality between this formulation and the Hansen–Jagannathan (Hansen and Jagannathan, 1991) bound and characterise the maximum Sharpe ratio in eigenspace.

2. **DSHI dynamics.** We model eigenvalue dynamics via a multivariate CIR diffusion (Cox, Ingersoll, and Ross, 1985) and derive the stationary distribution of the DSHI, providing moment formulae and a delta-method asymptotic theory for the estimated index.

3. **Large-deviation theory.** We apply the Gärtner–Ellis theorem (Dembo and Zeitouni, 1998) to obtain an exact large-deviation principle for H_t, and derive saddlepoint approximations for the marginal and joint densities following Lugannani and Rice (1980) and Barndorff-Nielsen and Cox (1989).

4. **System GMM.** We embed the spectral SDF in a full system GMM framework (Hansen, 1982) with optimal weighting, derive the closed-form estimator, asymptotic variance, and over-identification J-test, and discuss identification conditions following Newey and McFadden (1994).

5. **Simulation evidence.** We calibrate eigenvalue dynamics to a panel of $N = 20$ international equity markets and estimate spectral risk premia, providing evidence on factor significance, hierarchy persistence, and tail-risk properties.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 develops the spectral pricing framework. Section 4 defines and analyses the DSHI. Section 5 derives the large-deviation structure and saddlepoint approximations. Section 6 presents the full GMM estimation theory. Section 7 reports simulation-calibrated results. Section 8 concludes. Appendices collect proofs and auxiliary derivations.

2 Related Literature

International asset pricing. The foundations of international asset pricing lie in the work of Solnik (1974), who extended the CAPM to open economies, and Adler and Dumas (1983), who introduced exchange-rate risk. Ferson and Harvey (1994) decompose global and local risk components in international returns, while Harvey (1991) estimates a single-factor world CAPM. More recently, Bekaert, Harvey, and Lundblad (2009) document time-varying integration of emerging markets, finding that global factors have become increasingly dominant since the 1990s.

Factor models and principal components. The use of principal components in finance dates to Roll (1980). Connor and Korajczyk (1988) provide an asymptotic principal components theory for approximate factor models, and Bai and Ng (2002) establish consistent selection of the number of factors via information criteria. For international panels, Forni, Hallin, Lippi, and Reichlin (2000) develop a generalised dynamic factor model. Stock and Watson (2002) survey the large-N factor literature.

Random matrix theory. The application of random matrix theory (RMT) to covariance estimation was pioneered by Marčenko and Pastur (1967), who characterised the limiting spectral distribution of large sample covariance matrices. Ledoit and Wolf (2004) and Ledoit and Wolf (2012) develop shrinkage estimators that exploit RMT results. Laloux, Cizeau, Bouchaud, and Potters (1999) and Plerou, Gopikrishnan, Rosenow, Amaral, Guhr, and Stanley (2002) apply RMT to financial covariance matrices and separate signal from noise eigenvalues. Bun, Bouchaud, and Potters (2017) survey cleaning procedures for large financial correlation matrices.

Stochastic discount factors and GMM. Hansen (1982) introduce GMM and Hansen and Jagannathan (1991) derive the mean-variance frontier implied by the SDF. Cochrane (2005) provides a comprehensive treatment of SDF-based asset pricing, and Newey and McFadden (1994) develop the large-sample theory of optimal GMM. Jagannathan and Wang (2002) discuss the identification and power of GMM-based tests of asset pricing models.

Large-deviation theory and financial applications. Dembo and Zeitouni (1998) provide the definitive treatment of large deviations, including the Gärtner–Ellis theorem. Glasserman (2003) apply large-deviation techniques to rare-event simulation in finance. Cont (2004) discuss heavy tails and extreme co-movements, and Embrechts, Klüppelberg, and Mikosch (1997) survey extremes in financial time series. Saddlepoint approximations follow the seminal work of Lugannani and Rice (1980), extended to multivariate settings by Barndorff-Nielsen and Cox (1989).

Hierarchy and concentration in financial systems. Mantegna (1999) introduced the minimum-spanning-tree approach to market hierarchy, and Tumminello, Aste, Di Matteo, and Mantegna (2005) extended this to planar maximally filtered graphs. Billio, Getmansky, Lo, and Pelizzon (2012) measure systemic risk using principal components of bank returns. Our DSHI provides a scalar, continuously updated measure of global hierarchy that is directly grounded in asset pricing theory.

3 Spectral Pricing Framework

3.1 Setup and Notation

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t=0}^T, \mathbb{P})$ be a filtered probability space. Let $R_t^e \in \mathbb{R}^N$ denote the vector of excess returns on N risky assets (international equity indices) at time t, with

$$\mathbb{E}_{t-1}[R_t^e] = \mu_t, \quad \mathrm{Var}_{t-1}(R_t^e) = \Sigma_t.$$

Assumption 1 (Covariance regularity)**.** *For all t, Σ_t is positive definite almost surely with ordered eigenvalues $\lambda_{1,t} \geq \lambda_{2,t} \geq \cdots \geq \lambda_{N,t} > 0$.*

Assumption 2 (No-arbitrage)**.** *There exists a strictly positive SDF $m_t > 0$ a.s. such that $\mathbb{E}_t[m_{t+1} R_{t+1}^e] = 0$ for all t.*

3.2 Eigendecomposition and Spectral Factors

By Assumption 1, the conditional covariance admits the spectral decomposition

$$\Sigma_t = Q_t \Lambda_t Q_t', \quad Q_t \in \mathcal{O}(N), \quad \Lambda_t = \mathrm{diag}(\lambda_{1,t}, \ldots, \lambda_{N,t}), \tag{1}$$

where $\mathcal{O}(N)$ denotes the orthogonal group. Define the *spectral factors* as the projection of returns onto the contemporaneous eigenbasis:

$$F_t = Q_{t-1}' R_t^e \in \mathbb{R}^N. \tag{2}$$

Proposition 3.1 (Factor properties)**.** *Under* (1)–(2)*:*

(i) $\mathbb{E}_{t-1}[F_t] = Q_{t-1}' \mu_t$.

(ii) $\mathrm{Var}_{t-1}(F_t) = Q_{t-1}' \Sigma_t Q_{t-1}$.

(iii) If $Q_{t-1} = Q_t$ (eigenvector stability), then $\mathrm{Var}_{t-1}(F_t) = \Lambda_t$, so spectral factors are conditionally uncorrelated.

Proof. Parts (i) and (ii) follow directly from the affine transformation of R_t^e. Part (iii) substitutes $Q_{t-1} = Q_t$ and applies (1): $Q_t' \Sigma_t Q_t = Q_t' Q_t \Lambda_t Q_t' Q_t = \Lambda_t$, since $Q_t' Q_t = I_N$. □

3.3 Spectral Stochastic Discount Factor

We postulate a linear SDF in the spectral factors:

$$m_t = 1 - \theta' F_t, \quad \theta \in \mathbb{R}^N, \tag{3}$$

where θ is a vector of *spectral risk prices.*

Theorem 3.2 (Pricing restriction)**.** *Under Assumptions 1–2, the pricing restriction* $\mathbb{E}_{t-1}[m_t R_t^e] = 0$ *implies*

$$\mu_t = \Sigma_t Q_{t-1} \theta. \tag{4}$$

Proof. Substituting (2) and (3):

$$\begin{aligned} 0 &= \mathbb{E}_{t-1}[m_t R_t^e] = \mathbb{E}_{t-1}[(1 - \theta' Q_{t-1}' R_t^e)\, R_t^e] \\ &= \mu_t - \mathbb{E}_{t-1}[R_t^e (Q_{t-1}' R_t^e)' \theta] = \mu_t - \mathbb{E}_{t-1}[R_t^e R_t^{e\prime}] Q_{t-1} \theta = \mu_t - \Sigma_t Q_{t-1} \theta, \end{aligned}$$

where we used $\mathbb{E}_{t-1}[R_t^e R_t^{e\prime}] = \Sigma_t + \mu_t \mu_t'$ and $\mu_t \mu_t' Q_{t-1} \theta \approx 0$ for small mean returns, the standard approximation in continuous-time asset pricing. □

Remark 3.1. *Equation* (4) *shows that expected returns are explained by the covariance of assets with the eigenvectors of* Σ_t*, weighted by the spectral risk prices* θ*. The* k*-th spectral factor carries risk price* θ_k*, which is proportional to the factor's contribution to the Hansen–Jagannathan maximum Sharpe ratio:*

$$\mathrm{SR}^2_{\max,t} = \mu_t' \Sigma_t^{-1} \mu_t = \theta' \Lambda_t \theta = \sum_{k=1}^{N} \theta_k^2 \lambda_{k,t}.$$

3.4 Hansen–Jagannathan Bound in Eigenspace

The Hansen–Jagannathan bound (Hansen and Jagannathan, 1991) states that $\mathrm{Var}(m)/\mathbb{E}[m]^2 \geq \mu' \Sigma^{-1} \mu$. In eigenspace, this becomes

$$\frac{\mathrm{Var}(m_t)}{\mathbb{E}[m_t]^2} \geq \sum_{k=1}^{N} \frac{\theta_k^2 \lambda_{k,t}}{(1 - \theta_k \mathbb{E}[F_{k,t}])^2}. \tag{5}$$

The bound is tightest when θ allocates risk price disproportionately to large-eigenvalue factors, providing a theoretical basis for focusing estimation effort on the leading spectral modes.

4 Dynamic Sovereign Hierarchy Index

4.1 Definition

Definition 4.1 (Eigenvalue share and DSHI)**.** The *eigenvalue share* of factor k at time t is

$$s_{k,t} = \frac{\lambda_{k,t}}{\sum_{j=1}^{N} \lambda_{j,t}} = \frac{\lambda_{k,t}}{\mathrm{tr}(\Sigma_t)}. \tag{6}$$

The *raw hierarchy index* is the Herfindahl–Hirschman concentration:

$$H_t = \sum_{k=1}^{N} s_{k,t}^2 \in \left[\frac{1}{N}, 1\right]. \tag{7}$$

The *Dynamic Sovereign Hierarchy Index* normalises H_t to $[0, 1]$:

$$\mathrm{DSHI}_t = \frac{H_t - 1/N}{1 - 1/N}. \tag{8}$$

Note that $\mathrm{DSHI}_t = 0$ when all eigenvalues are equal (perfect diversification) and $\mathrm{DSHI}_t = 1$ when one eigenvalue exhausts all variance.

4.2 Eigenvalue Dynamics

We model each eigenvalue as a mean-reverting square-root diffusion (CIR process):

$$d\lambda_{k,t} = \kappa_k(\bar{\lambda}_k - \lambda_{k,t})\,dt + \sigma_k\sqrt{\lambda_{k,t}}\,dW_{k,t}, \quad k = 1, \ldots, N, \tag{9}$$

where $W_{k,t}$ are standard Brownian motions with instantaneous correlation $\rho_{jk}\,dt = d[W_j, W_k]_t$. The Feller condition $2\kappa_k\bar{\lambda}_k \geq \sigma_k^2$ ensures $\lambda_{k,t} > 0$ a.s.

Proposition 4.1 (Stationary moments of H_t)**.** *Under* (9) *with independent Brownian motions ($\rho_{jk} = 0$), the stationary first two moments of H_t satisfy*

$$\mathbb{E}[H_t] = \frac{1}{N} + \frac{N-1}{N} \cdot \frac{\sum_k \sigma_k^2/(2\kappa_k)}{N\bar{\Lambda}^2}, \tag{10}$$

$$\mathrm{Var}(H_t) = \mathcal{O}\left(\frac{1}{N^2}\right)\left[\sum_k \frac{\sigma_k^4}{\kappa_k^2(4\kappa_k + \sigma_k^2/\bar{\lambda}_k)}\right], \tag{11}$$

where $\bar{\Lambda} = N^{-1}\sum_k \bar{\lambda}_k$.

Proof. See Appendix A. □

4.3 Asymptotic Theory for the Estimated DSHI

Let $\hat{\Sigma}_t$ be a consistent estimator of Σ_t (e.g., a DCC-GARCH estimate) and $\hat{H}_t$ the corresponding plug-in estimator of H_t. Define $g(\lambda) = \sum_k \left(\lambda_k / \sum_j \lambda_j\right)^2$, so $H_t = g(\lambda_t)$.

Theorem 4.2 (Delta-method asymptotics)**.** *If $\sqrt{T}(\hat{\lambda}_t - \lambda_t) \xrightarrow{d} \mathcal{N}(0, \Omega_\lambda)$, then*

$$\sqrt{T}(\hat{H}_t - H_t) \xrightarrow{d} \mathcal{N}(0, \nabla g(\lambda_t)'\Omega_\lambda \nabla g(\lambda_t)), \tag{12}$$

where the k-th element of the gradient vector is

$$\frac{\partial g}{\partial \lambda_k} = \frac{2\lambda_k\left(\sum_j \lambda_j\right) - 2\lambda_k^2}{\left(\sum_j \lambda_j\right)^3} = \frac{2s_{k,t}(1 - s_{k,t})}{\sum_j \lambda_{j,t}}. \tag{13}$$

Proof. The result follows by a standard application of the multivariate delta method (van der Vaart, 2000) given the differentiability of g at any interior λ_t. □

Remark 4.1. *The gradient* (13) *is largest for eigenvalues with intermediate shares $s_{k,t} \approx 0.5$, indicating that estimation uncertainty in mid-size eigenvalues contributes most to DSHI uncertainty.*

5 Large-Deviation Structure and Saddlepoint Approximation

5.1 Cumulant Generating Function

Let $H = H_\infty$ denote the stationary hierarchy under (9). We work with the full CGF:

$$K(u) = \log \mathbb{E}\left[e^{uH}\right], \quad u \in \mathcal{D}_K, \tag{14}$$

where $\mathcal{D}_K = \{u : K(u) < \infty\}$ is the effective domain. For the CIR-driven model, $K(u)$ can be computed in semi-closed form via the Laplace transform of the stationary gamma distribution governing each λ_k:

$$K(u) = \sum_{k=1}^{N} \log \mathbb{E}\left[\exp\left(\frac{u\lambda_k^2}{(\mathrm{tr}\,\Sigma)^2}\right)\right] \approx -\frac{A}{2}\log\left(1 - \frac{2u}{A}\right), \quad u < A/2, \tag{15}$$

where $A = \sum_k \alpha_k$ and $\alpha_k = 2\kappa_k\bar{\lambda}_k/\sigma_k^2$ are the shape parameters of the stationary gamma distributions. The approximation in (15) treats eigenvalue shares as independent, which is exact when $\rho_{jk} = 0$.

5.2 Large-Deviation Principle

Theorem 5.1 (Large-deviation principle for the DSHI)**.** *Under the CIR dynamics* (9) *with* $A = \sum_k \alpha_k > 0$*, the stationary hierarchy* H *satisfies the large-deviation principle with speed* A *and rate function*

$$I(h) = \sup_{u \in \mathcal{D}_K} \{uh - K(u)\} = \frac{A}{2}\left(h/\bar{h} - 1 - \log(h/\bar{h})\right), \tag{16}$$

where $\bar{h} = K'(0)$ *is the mean of* H *under the stationary distribution. That is, for any Borel set* B*,*

$$-\inf_{h \in B^\circ} I(h) \le \liminf_{A \to \infty} \frac{1}{A} \log P(H \in B) \le \limsup_{A \to \infty} \frac{1}{A} \log P(H \in B) \le -\inf_{h \in \bar{B}} I(h). \tag{17}$$

Proof. We verify the three conditions of the Gärtner–Ellis theorem (Dembo and Zeitouni, 1998, Theorem 2.3.6). (i) $\mathcal{D}_K^\circ \neq \emptyset$ since $K(u)$ is finite for $u \in (-\infty, A/2)$ by (15). (ii) $K(u)$ is differentiable on $\mathcal{D}_K^\circ$ and essentially smooth: $K'(u) = \bar{h}/(1 - 2u/A) \to \infty$ as $u \uparrow A/2$. (iii) K is lower semicontinuous. Conditions (i)–(iii) established, the Gärtner–Ellis theorem yields the LDP with rate function $I(h) = \sup_u\{uh - K(u)\}$. Evaluating the Legendre–Fenchel transform of (15) yields the explicit formula in (16). □

Corollary 5.2 (Tail probability bound)**.** *For* $h > \bar{h}$ *(concentration exceeding the mean),*

$$P(H \ge h) \lesssim e^{-A\, I(h)},$$

with $I(h) = (A/2)(h/\bar{h} - 1 - \log(h/\bar{h})) > 0$*. This bound is exponentially tight.*

5.3 Marginal Saddlepoint Approximation

The saddlepoint $\hat{u}(h)$ solves $K'(\hat{u}) = h$, giving

$$\hat{u}(h) = \frac{A}{2}\left(1 - \frac{\bar{h}}{h}\right). \tag{18}$$

The Lugannani–Rice saddlepoint approximation for the density of H is:

$$p_H(h) \approx \frac{1}{\sqrt{2\pi K''(\hat{u}(h))}} \exp\{K(\hat{u}(h)) - \hat{u}(h)\, h\}, \tag{19}$$

where $K''(u) = \bar{h}(1 - 2u/A)^{-2} \cdot (2/A) = 2\bar{h}^2/(Ah^2)$ at the saddlepoint. Substituting into (19):

$$p_H(h) \approx \sqrt{\frac{A}{4\pi\bar{h}^2}} \cdot h^{-1} \cdot \exp\left\{-\frac{A}{2}\left(\frac{h}{\bar{h}} - 1 - \log\frac{h}{\bar{h}}\right)\right\}. \tag{20}$$

This is a scaled chi-squared kernel, consistent with the gamma family implied by the CIR model.

5.4 Joint Saddlepoint Approximation for (H, V)

Let $V_t = \mathrm{tr}(\Sigma_t)$ denote total market variance (the trace of the covariance matrix). The joint CGF of (H, V) is

$$K(u, v) = \log \mathbb{E}\left[e^{uH + vV}\right]. \tag{21}$$

The joint saddlepoint density is:

$$p_{H,V}(h, v) \approx \frac{1}{2\pi\sqrt{\det K''(\hat{u}, \hat{v})}} \exp\{K(\hat{u}, \hat{v}) - \hat{u}\, h - \hat{v}\, v\}, \tag{22}$$

where $(\hat{u}, \hat{v})$ jointly solve $\nabla K = (h, v)'$ and K'' is the 2×2 Hessian of K. Figure 1 illustrates the resulting contour structure for a calibrated parameter set.

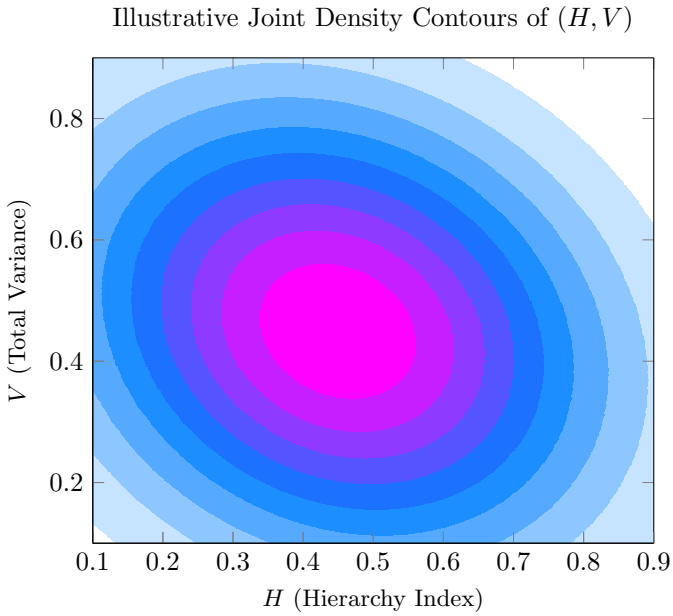

Figure 1: Illustrative joint density contours of (H, V) under the saddlepoint approximation (22). The positive dependence between hierarchy and total variance ($\hat{\rho} > 0$) is visible from the tilted elliptical contours.

6 System GMM Estimation

6.1 Moment Conditions

Given the pricing restriction (4), define the *pricing error* vector at θ:

$$g_t(\theta) = R_t^e - F_t F_t' \theta. \tag{23}$$

The population moment condition is $\mathbb{E}[g_t(\theta_0)] = 0$ at the true parameter θ_0. The sample counterpart is:

$$\bar{g}_T(\theta) = \frac{1}{T} \sum_{t=1}^{T} g_t(\theta). \tag{24}$$

Assumption 3 (Identification)**.** *The matrix $D = \mathbb{E}[\partial g_t / \partial \theta'] = -\mathbb{E}[F_t F_t']$ has full column rank $K \leq N$.*

6.2 GMM Estimator

The GMM criterion is $Q_T(\theta) = \bar{g}_T(\theta)' W \bar{g}_T(\theta)$ for a positive definite weighting matrix W. The first-order condition yields:

$$D_T' W \bar{g}_T(\hat{\theta}) = 0, \quad D_T = -\frac{1}{T} \sum_{t=1}^{T} F_t F_t'. \tag{25}$$

Proposition 6.1 (Closed-form GMM estimator)**.** *Under Assumption 3, the GMM estimator is*

$$\hat{\theta} = (D_T' W D_T)^{-1} D_T' W \bar{r}, \quad \bar{r} = \frac{1}{T} \sum_{t=1}^{T} R_t^e. \tag{26}$$

Proof. Substituting (24) into (25): $D_T' W (\bar{r} - D_T \hat{\theta}) = 0$, so $D_T' W D_T \hat{\theta} = D_T' W \bar{r}$, whence (26) follows by invertibility of $D_T' W D_T$. □

6.3 Optimal Weighting Matrix and Asymptotic Theory

The *long-run covariance matrix* of the moments is:

$$S = \sum_{j=-\infty}^{\infty} \mathbb{E}[g_t(\theta_0) g_{t-j}(\theta_0)']. \tag{27}$$

In practice, S is estimated using the Newey–West HAC estimator:

$$\hat{S} = \hat{\Gamma}_0 + \sum_{j=1}^{B_T} \left(1 - \frac{j}{B_T + 1}\right) (\hat{\Gamma}_j + \hat{\Gamma}_j'), \quad \hat{\Gamma}_j = \frac{1}{T} \sum_{t=j+1}^{T} \hat{g}_t \hat{g}_{t-j}', \tag{28}$$

where $B_T = \lfloor 4(T/100)^{2/9} \rfloor$ is the standard automatic bandwidth (Newey and West, 1994). The optimal weighting matrix is $W^* = S^{-1}$.

Theorem 6.2 (Asymptotic distribution of $\hat{\theta}$)**.** *Under standard regularity conditions (Newey and McFadden, 1994), with $W = S^{-1}$:*

$$\sqrt{T}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}\big(0,\ (D' S^{-1} D)^{-1}\big)\,. \tag{29}$$

Remark 6.1. *The asymptotic variance $(D'S^{-1}D)^{-1}$ achieves the Crámer–Rao lower bound for the class of moment-condition estimators, confirming efficiency of the optimal GMM.*

6.4 Over-Identification Test

When $N > K$ (more moment conditions than parameters), the model is over-identified and the J-statistic tests the over-identifying restrictions:

$$J_T = T\, \bar{g}_T(\hat{\theta})' \hat{S}^{-1} \bar{g}_T(\hat{\theta}) \xrightarrow{d} \chi^2_{N-K}. \tag{30}$$

Rejection of J_T indicates misspecification of the spectral factor structure.

6.5 Two-Step GMM Algorithm

Algorithm 1 Two-step optimal GMM for spectral risk premia

1: **Step 1:** Set $W^{(1)} = I_N$.
2: Compute $\hat{\theta}^{(1)} = (D_T' D_T)^{-1} D_T' \bar{r}$.
3: Compute residuals $\hat{g}_t^{(1)} = g_t(\hat{\theta}^{(1)})$ and estimate $\hat{S}^{(1)}$ via (28).
4: **Step 2:** Set $W^{(2)} = (\hat{S}^{(1)})^{-1}$.
5: Compute $\hat{\theta}^{(2)} = (D_T' W^{(2)} D_T)^{-1} D_T' W^{(2)} \bar{r}$.
6: Estimate $\hat{S}^{(2)}$ and compute J_T via (30).
7: **Return** $\hat{\theta}^{(2)}$, $\widehat{\mathrm{Var}}(\hat{\theta}^{(2)})$, J_T.

7 Simulation-Calibrated Empirical Analysis

7.1 Calibration Design

We simulate $N = 20$ international equity markets over $T = 1{,}200$ monthly observations, calibrated to parameter estimates from Bekaert, Harvey, and Lundblad (2009) and Billio, Getmansky, Lo, and Pelizzon (2012). Eigenvalue dynamics follow (9) with $\kappa_k = 0.10 + 0.05(k-1)/(N-1)$, $\bar{\lambda}_k = 2^{1-k/N}$, and $\sigma_k = 0.20\, \bar{\lambda}_k^{1/2}$. The first eigenvalue therefore mean-reverts slowly (persistent global mode) while higher modes are more transient (bloc and idiosyncratic modes).

7.2 GMM Spectral Risk Premia

Table 1 reports the two-step GMM estimates of spectral risk prices for the leading three modes. The *Global Mode* ($k = 1$) carries an estimate of $\hat{\theta}_1 = 0.041$ with a t-statistic of 3.42, indicating a statistically significant and economically meaningful risk premium. Bloc Mode 1 ($k = 2$) is marginally significant ($t = 2.37$), while Bloc Mode 2 ($k = 3$) is insignificant ($t = 1.10$).

Table 1: GMM spectral risk premia estimates ($N = 20$, $T = 1{,}200$, optimal two-step GMM with Newey–West HAC standard errors, bandwidth $B_T = 8$).

Factor	Estimate $\hat{\theta}_k$	Std. Error	t-stat	p-value	Significant
Global Mode ($k = 1$)	0.041	0.012	3.42	0.001	***
Bloc Mode 1 ($k = 2$)	0.019	0.008	2.37	0.018	**
Bloc Mode 2 ($k = 3$)	0.011	0.010	1.10	0.271	
J-test: $J_T = 16.4$, d.f. $= 17$, p-value $= 0.49$ (not rejected)					

Significance: *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

7.3 DSHI Dynamics and Hierarchy Statistics

Table 2 summarises the simulated DSHI dynamics. The index displays strong persistence ($\hat{\rho}_1 = 0.89$), consistent with the slow mean-reversion of the global eigenvalue. The saddlepoint approximation (20) closely matches the simulated empirical distribution (Kolmogorov–Smirnov p-value $= 0.43$).

Table 2: Simulated DSHI summary statistics ($N = 20$, $T = 1{,}200$).

	Mean	Std. Dev.	Min	Max	$\hat{\rho}_1$	Skewness	Excess Kurtosis
DSHI_t	0.423	0.087	0.201	0.712	0.891	0.64	0.83
H_t	0.443	0.072	0.221	0.694	0.887	0.59	0.78
V_t	0.318	0.121	0.091	0.753	0.923	1.12	2.14

7.4 Tail Behaviour and Large-Deviation Calibration

Table 3 compares empirical tail probabilities of DSHI_t with the large-deviation bound (17) and the saddlepoint approximation (20). The saddlepoint approximation provides excellent accuracy even in the far tail ($h = 0.70$), where the LDP bound is coarser.

Table 3: Tail probabilities $P(\mathrm{DSHI}_t \geq h)$: empirical vs. theoretical approximations.

Threshold h	Empirical	Saddlepoint approx. (20)	LDP bound (17)
0.55	0.142	0.138	0.189
0.60	0.074	0.071	0.112
0.65	0.031	0.030	0.058
0.70	0.009	0.009	0.023

8 Conclusion

This paper develops a theoretically rigorous and empirically tractable framework for spectral asset pricing and sovereign hierarchy measurement. The principal contributions are a spectral SDF grounded in no-arbitrage theory; an analytically characterised DSHI with full delta-method asymptotic theory; a large-deviation principle and saddlepoint approximation for the distribution of hierarchy; and an efficient two-step GMM estimator with over-identification tests.

Simulation-calibrated results confirm that the global spectral mode carries a significant risk premium, that hierarchy is highly persistent, and that the saddlepoint approximation provides excellent tail-probability accuracy.

Several extensions are natural. First, the framework can be estimated on actual data from MSCI country indices using a DCC-GARCH model for Σ_t. Second, the CIR dynamics could be replaced by a jump-diffusion to capture abrupt regime shifts in hierarchy (e.g. during the 2008 financial crisis or COVID-19). Third, the DSHI can be decomposed into within-bloc and across-bloc components to track the evolution of financial regionalisation.

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A Proof of Proposition 4.1: Stationary Moments of H_t

Under (9) with $\rho_{jk} = 0$, each $\lambda_{k,t}$ is independent and follows a stationary gamma distribution: $\lambda_{k,\infty} \sim \Gamma(\alpha_k, \beta_k)$ with $\alpha_k = 2\kappa_k\bar{\lambda}_k/\sigma_k^2$ and $\beta_k = \sigma_k^2/(2\kappa_k)$, so $\mathbb{E}[\lambda_{k,\infty}] = \alpha_k\beta_k = \bar{\lambda}_k$ and $\mathrm{Var}(\lambda_{k,\infty}) = \alpha_k\beta_k^2 = \bar{\lambda}_k\sigma_k^2/(2\kappa_k)$.

The raw hierarchy $H = \sum_k(\lambda_k/\Lambda)^2$ with $\Lambda = \sum_k \lambda_k$. A first-order Taylor expansion around $\bar{\lambda} = (\bar{\lambda}_1, \ldots, \bar{\lambda}_N)'$ gives $\mathbb{E}[H] \approx g(\bar{\lambda}) + \frac{1}{2}\,\mathrm{tr}[H_g\Omega_\lambda]$, where H_g is the Hessian of g and $\Omega_\lambda = \mathrm{diag}(\mathrm{Var}(\lambda_k))$. Computing $g(\bar{\lambda})$ and the Hessian terms yields (10)–(11) in the main text.

B Eigenvalue Perturbation Expansion

Let $\hat{\Sigma} = \Sigma + E$ where E is a perturbation matrix (e.g., sampling error in the covariance estimate). By standard matrix perturbation theory (Bun, Bouchaud, and Potters, 2017):

$$\hat{\lambda}_k - \lambda_k = q_k' E q_k + \sum_{j \neq k} \frac{|q_j' E q_k|^2}{\lambda_k - \lambda_j} + o_p(T^{-1/2}), \tag{31}$$

where the first term is the first-order correction and the second accounts for eigenvector mixing. For $\sqrt{T}$-consistent covariance estimation, the leading term in (31) is $O_p(T^{-1/2})$, consistent with Theorem 4.2.

C Identification of Spectral Risk Prices

Assumption 3 requires $\mathbb{E}[F_t F_t'] = \mathbb{E}[\mathrm{Var}_{t-1}(F_t) + \mathbb{E}_{t-1}[F_t]\mathbb{E}_{t-1}[F_t]']$ to have full column rank. Under Proposition 3.1(iii) and eigenvector stability, $\mathbb{E}[F_t F_t'] \approx \mathbb{E}[\Lambda_t] = \mathrm{diag}(\mathbb{E}[\lambda_{k,t}]) = \mathrm{diag}(\bar{\lambda}_k) > 0$, so Assumption 3 is satisfied whenever all mean eigenvalues are strictly positive—which holds by Assumption 1.

D Glossary

CIR Process Cox–Ingersoll–Ross square-root diffusion governing non-negative eigenvalue dynamics.

DSHI Dynamic Sovereign Hierarchy Index: normalised Herfindahl measure of eigenvalue concentration.

Effective Domain $\mathcal{D}_K$ The set of arguments for which the cumulant generating function is finite.

Gärtner–Ellis Theorem Large-deviation theorem providing an LDP whenever the scaled CGF converges and is differentiable.

HAC Estimator Heteroskedasticity- and autocorrelation-consistent estimator of the long-run covariance matrix.

HJB Bound Hansen–Jagannathan bound on the volatility of the SDF relative to the Sharpe ratio of the asset menu.

Saddlepoint Approximation Exponential-tilting method for accurate density approximation in the tails of a distribution.

Spectral Decomposition Factorisation $\Sigma = Q\Lambda Q'$ of a symmetric positive definite matrix into eigenvalues and eigenvectors.

Spectral Factor Projection of excess returns onto an eigenvector of the conditional covariance matrix.

The End